We create chemistry

# Safety data sheet



## 1. Identification

### Product identifier

## Aviso DF

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, fungicide

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Skin Sens. 1B
Repr. 2 (fertility)
Repr. 2 (unborn child)
STOT RE (Skeletal muscle) 2

Aquatic Acute 1
Aquatic Chronic 1

For the classifications not written out in full in this section the full text can be found in section 16.

## Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

| | |
|---|---|
| H317 | May cause an allergic skin reaction. |
| H361 | Suspected of damaging fertility. Suspected of damaging the unborn child. |
| H373 | May cause damage to organs (Skeletal muscle) through prolonged or repeated exposure. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P260 | Do not breathe dust/gas/mist/vapours. |
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P272 | Contaminated work clothing should not be allowed out of the workplace. |

Precautionary Statements (Response):

| | |
|---|---|
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P302 + P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P333 + P313 | If skin irritation or rash occurs: Get medical attention. |
| P391 | Collect spillage. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |

Precautionary Statements (Storage):

P405 Store locked up.

Precautionary Statements (Disposal):
P501 Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide, metiram

### Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

crop protection product, fungicide, water dispersible granules

Hazardous ingredients (GHS)
According to UN GHS criteria

metiram

| | |
|---|---|
| Content (W/W): 57 % | Skin Sens. 1 |
| CAS Number: 9006-42-2 | STOT RE (Skeletal muscle) 2 |
| | Aquatic Acute 1 |
| | Aquatic Chronic 1 |
| | H317, H373, H400, H410 |

cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide

| | |
|---|---|
| Content (W/W): 4,8 %<br>CAS Number: 57966-95-7<br>EC-Number: 261-043-0 | Acute Tox. 4 (oral)<br>Skin Sens. 1<br>Repr. 2 (fertility)<br>Repr. 2 (unborn child)<br>STOT RE (Blood, Thymus gland) 2<br>Aquatic Acute 1<br>Aquatic Chronic 1<br>H302, H317, H373, H361, H400, H410 |

Sodium diisobutylnaphthalenesulphonate

| | |
|---|---|
| Content (W/W): < 3 %<br>CAS Number: 27213-90-7<br>EC-Number: 248-326-4 | Acute Tox. 4 (Inhalation - dust)<br>Acute Tox. 4 (oral)<br>Eye Dam./Irrit. 1<br>STOT SE 3 (irr. to respiratory syst.)<br>Aquatic Acute 3<br>Aquatic Chronic 3<br>H318, H335, H302 + H332, H402, H412 |

Lignosulfonic acid, sodium salt

Content (W/W): < 40 %
CAS Number: 8061-51-6

Sodium sulphate

Content (W/W): < 5 %
CAS Number: 7757-82-6
EC-Number: 231-820-9

Glucose

Content (W/W): < 5 %
CAS Number: 50-99-7
EC-Number: 200-075-1

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### Most important symptoms and effects, both acute and delayed
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

### Indication of any immediate medical attention and special treatment needed
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

### Extinguishing media
Suitable extinguishing media:
dry powder, foam, water spray

Unsuitable extinguishing media for safety reasons:
carbon dioxide

### Special hazards arising from the substance or mixture
Carbon monoxide, Carbon dioxide, Hydrogen chloride, nitrogen oxides, cyanides, sulfur oxides, halogenated compounds
The substances/groups of substances mentioned can be released in case of fire.

### Advice for fire-fighters
Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

### Personal precautions, protective equipment and emergency procedures
Avoid dust formation. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

### Environmental precautions
Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

### Methods and material for containment and cleaning up
For small amounts: Contain with dust binding material and dispose of.
For large amounts: Sweep/shovel up.
Avoid raising dust. Collect waste in suitable containers, which can be labeled and sealed. Dispose of absorbed material in accordance with regulations. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

### Precautions for safe handling
No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Avoid dust formation. Dust can form an explosive mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

### Conditions for safe storage, including any incompatibilities
Segregate from foods and animal feeds.
Further information on storage conditions: Protect against moisture. Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months
Keep container dry.

### Specific end use(s)
For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

50-99-7: Glucose
7757-82-6: Sodium sulphate
9006-42-2: metiram
8061-51-6: Lignosulfonic acid, sodium salt

### Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with high efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P3 or FFP3).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

### Information on basic physical and chemical properties

| | | |
|---|---|---|
| Form: | solid | |
| Colour: | brown | |
| Odour: | faint odour, smoky | |
| Odour threshold: | Not determined due to potential health hazard by inhalation. | |
| pH value: | approx. 4,5 - 7<br>(10 g/l) | |
| Melting point: | The product has not been tested. | |
| Boiling point: | The product has not been tested. | |
| Flash point: | not applicable, the product is a solid | |
| Evaporation rate: | not applicable | |
| Flammability: | not highly flammable | (Directive 84/449/EEC, A.10) |
| Lower explosion limit: | 150 g/m3<br>(50 °C) | (air) |

Upper explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.

Vapour pressure:
The product has not been tested.

Density: approx. 1,43 g/cm3 (OECD Guideline 109)
(20 °C)

Relative vapour density (air):
not applicable

Solubility in water: dispersible

Partitioning coefficient n-octanol/water (log Kow):
The statements are based on the properties of the individual components.

*Information on: metiram*
*Partitioning coefficient n-octanol/water (log Kow): 0,33*
*(pH value: 7)*
*1,9*
*(pH value: 5)*
*-0,37*
*(pH value: 9)*

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Partitioning coefficient n-octanol/water (log Kow): 0,64*
*(20 °C)*

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Self ignition: not self-igniting

Thermal decomposition: 145 °C, 20 kJ/kg (DSC (OECD 113))
180 °C, 160 kJ/kg (DSC (OECD 113))
340 °C, 50 kJ/kg (DSC (OECD 113))
455 °C, > 40 kJ/kg (DSC (OECD 113))
Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.

Viscosity, dynamic:
not applicable, the product is a solid

Explosion hazard: not explosive

Fire promoting properties: not fire-propagating (Directive 92/69/EEC, A.17)

## Other information

Self heating ability: It is not a substance capable of spontaneous heating.

## 10. Stability and Reactivity

### Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

### Chemical stability
The product is stable if stored and handled as prescribed/indicated.

### Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

### Conditions to avoid
See SDS section 7 - Handling and storage.

### Incompatible materials

Substances to avoid:
strong acids, strong oxidizing agents, strong bases

### Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Experimental/calculated data:
LD50 rat (oral): > 2.200 mg/kg (OECD Guideline 401)
No mortality was observed.

LC50 rat (by inhalation): > 5,2 mg/l 4 h (OECD Guideline 403)
No mortality was observed. An aerosol was tested.

LD50 rat (dermal): > 2.000 mg/kg (OECD Guideline 402)
No mortality was observed.

Irritation

Assessment of irritating effects:
Not irritating to the eyes. Not irritating to the skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
The product has not been tested. The statement has been derived from the properties of the individual components. Sensitization after skin contact possible.

*Information on: metiram*
*Experimental/calculated data:*
*Guinea pig maximization test guinea pig:* Caused skin sensitization in animal studies.

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Experimental/calculated data:*
*Guinea pig maximization test guinea pig:* sensitizing *(OECD Guideline 406)*
---

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of various animal studies gave no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Assessment of reproduction toxicity:*
*The results of animal studies suggest a fertility impairing effect.*
---

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Assessment of teratogenicity:*
*Indications of possible developmental toxicity/teratogenicity were seen in animal studies.*
---

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metiram*
*Assessment of repeated dose toxicity:*
*Damages the skeletal muscle.*

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Assessment of repeated dose toxicity:*
*The substance may damage the red blood cells after repeated ingestion of high doses, as shown in animal studies.*
---

Aspiration hazard

No aspiration hazard expected.
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.

Toxicity to fish:

LC50 (96 h) 0,452 mg/l, Oncorhynchus mykiss

Aquatic invertebrates:
EC50 (48 h) 1,17 mg/l, Daphnia magna

Aquatic plants:
EC50 (72 h) 0,114 mg/l, Pseudokirchneriella subcapitata

EC10 (72 h) 0,029 mg/l, Pseudokirchneriella subcapitata

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Chronic toxicity to fish:*
*No observed effect concentration (97 d) 0,00098 mg/l, Oncorhynchus mykiss (OECD Guideline 210)*
---

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Chronic toxicity to aquatic invertebrates:*
*No observed effect concentration (21 d) 0,067 mg/l, Daphnia magna (OECD Guideline 202, part 2)*
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## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metiram*
*Assessment biodegradation and elimination (H2O):*
*According to OECD criteria the product is not readily biodegradable but inherently biodegradable.*

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Assessment biodegradation and elimination (H2O):*
*Not readily biodegradable (by OECD criteria).*
---

## Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metiram*
*Bioaccumulation potential:*
*Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.*

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Bioaccumulation potential:*
*Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.*
---

### Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metiram*
*Assessment transport between environmental compartments:*
*Adsorption in soil: Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.*

*Information on: cymoxanil; 2-cyano-N-[(ethylamino)carbonyl]-2-(methoxyimino)acetamide*
*Assessment transport between environmental compartments:*
*Adsorption in soil: Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.*
---

### Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

### Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

### Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains METIRAM) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

RID

| | |
|---|---|
| UN number | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains METIRAM) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains METIRAM) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number: | UN 3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains METIRAM) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |

| | |
|---|---|
| | Marine pollutant: YES |
| Special precautions for user: | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number: | UN 3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (contains METIRAM) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

### Transport in bulk according to Annex II of MARPOL and the IBC Code

| | |
|---|---|
| Regulation: | Not evaluated |
| Shipment approved: | Not evaluated |
| Pollution name: | Not evaluated |
| Pollution category: | Not evaluated |
| Ship Type: | Not evaluated |

**Further information**
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 kg or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2:10.2.7; IATA: A197; TDS: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3 Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

## 16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:
Skin Sens. Skin sensitization

| | |
|---|---|
| Repr. | Reproductive toxicity |
| STOT RE | Specific target organ toxicity — repeated exposure |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| Acute Tox. | Acute toxicity |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| STOT SE | Specific target organ toxicity — single exposure |
| H317 | May cause an allergic skin reaction. |
| H373 | May cause damage to organs (Skeletal muscle) through prolonged or repeated exposure. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |
| H302 | Harmful if swallowed. |
| H361 | Suspected of damaging fertility. Suspected of damaging the unborn child. |
| H318 | Causes serious eye damage. |
| H335 | May cause respiratory irritation. |
| H302 + H332 | Harmful if swallowed or if inhaled |
| H402 | Harmful to aquatic life. |
| H412 | Harmful to aquatic life with long lasting effects. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.